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EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Class Test 2, Fall 2025 Semester

Course: CSE 110 Object Oriented Programming, Section 4
Instructor: Dr. Md. Hasanul Ferdaus, Assistant Professor, CSE Department
Full Marks: 20 (will be normalized out of 5)
Time: 20 Minutes

Note: There are TWO (2) questions; answer BOTH.

1. Create a superclass named Appliance with fields brand and powerRating (in watts). Include a method displayInfo() that prints these values. Create a subclass called WashingMachine that adds a field capacity (in kilograms) and a method showDetails() that prints its own data. Create a driver class called HomeService that contains the main() method. Within the main() method, create an object of WashingMachine and call both superclass and subclass methods to show the inherited behavior.

[CO2,
Mark: 10]

SAMPLE OUTPUT

Appliance: LG, 1500W
Capacity: 7 kg

ANSWER:

```
class Appliance {  
    String brand = "Apple";  
    int powerRating = 1500;
```

```
    Appliance(String b, int p) {  
        brand = b;  
        powerRating = p;
```

constructor?

```
    void displayInfo() {
```

```
        System.out.println("Appliance: " + brand + ", " +  
            powerRating + "W");
```

```
    }
```

```
}
```

2. Create a superclass named Alert that has a method trigger() which prints a generic alert message.
Create a subclass SMSAlert that overrides trigger() to print a more specific message related to sending an SMS.
Create a driver class named AlertSystem that contains the main() method. In the main() method, demonstrate dynamic method dispatch by storing an SMSAlert object in an Alert reference and calling trigger().

[CO2,
Mark: 10]

SAMPLE OUTPUT

Sending SMS alert...

ANSWER:

```
class Alert {
```

```
    void trigger() {
```

```
        System.out.println("Sending generic alert");
```

```
    }
```

```
}
```

```
class SMSAlert extends Alert {
```

```
    void trigger() {
```

```
        System.out.println("Sending sms alert.");
```

```
    }
```

```
}
```

```
public class AlertSystem {
```

```
    public static void main(String[] args) {
```

```
        Alert Alert msg = new SMSAlert();
```

```
        msg.trigger();
```

```
    } }
```